

BOUDROUSS RÉDA | CV

Software Engineer | OCaml, WebAssembly & Static Analysis

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Location: Paris, France

Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology) at Sorbonne University, specialized in **static analysis**, and **Web technologies**. I ported **MOPSA**, an OCaml/C/C++ abstract-interpretation analyzer, to **WebAssembly**, which took me deep into the OCaml **runtime**, its C FFI and float semantics on Wasm.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc

2026

- Compiled the **OCaml runtime** and the full native stack of **MOPSA** (Clang frontend, Apron, GMP, MPFR) into a single static **WebAssembly** module with Emscripten, an approach chosen over `js_of_ocaml` and `wasm_of_ocaml` because it brings the C/C++ half of the codebase along.
- Replaced the **runtime**'s dynamic FFI loading with a generated static table of **~1,400 C primitives**, and root-caused an unsigned-wraparound bug in the `Tag_val` macro, a native-vs-Wasm semantics mismatch that traps under Wasm's bounds-checked addressing.
- Addressed the **soundness** hazard of Wasm's environment, such as fixed rounding mode (no `fesetround`).
- Shipped a fully client-side **playground** (Web Workers around a non-reentrant runtime, blocking stdin over `SharedArrayBuffer` + `Atomics` for the interactive REPL and DAP modes) and a technical **write-up**.

Reactant analyzer - github.com/rboudrouss/reactant-analyzer

2026

- Designed and implemented an **abstract-interpretation**-based static analyzer in **Rust** (~30k lines, 500+ tests) that catches **React** hook bugs beyond the reach of syntactic linters, such as infinite render loops, derived state and unstable dependencies.
- Evaluates components in abstract domains (intervals, constant sets, reference-stability tags) over a dedicated CFG-based IR and flags an infinite render loop when the state fixpoint **widens**. Resolves and inlines custom hooks across files, with concrete semantics based on the React-tRace paper (OOPSLA 2025).

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe

2025 – 2026

- Designed complete **test strategies** (unit, integration, E2E) for legacy products with no prior coverage, reaching **~3,000 test cases** across 4 stacks (TypeScript, Python, PHP, JS), partly through AI-agent test production under versioned guardrails and systematic human review.
- Built objective **quality oracles** (VMAF, black-frame/freeze detection, audio analysis) with `ffmpeg` to validate a live video encoder.
- Built complete **GitLab CI/CD** pipelines and shared CI components (npm publishing, centralized Allure reporting) consumed by 4+ repositories.

Technical Lead, then President (2024–25) - Association Play Sorbonne Université

2023 – 2026

- Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking) and built a **local-first** mobile web app. As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University

2024 – 2026

- Coursework: compiler construction, design & implementation of **programming languages** and their runtimes (interpreters vs. compilers), formal semantics, **type systems** and **static analysis by abstract interpretation**, concurrent & distributed programming.

BSc in Computer Science - Sorbonne University

2020 – 2024

- Algorithms, computer architecture, operating systems, networks, databases and functional programming.